Best Practices in Data Governance, Preparation and Analytics (CA6003)

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

# Pitfalls in EDA- Paradoxes

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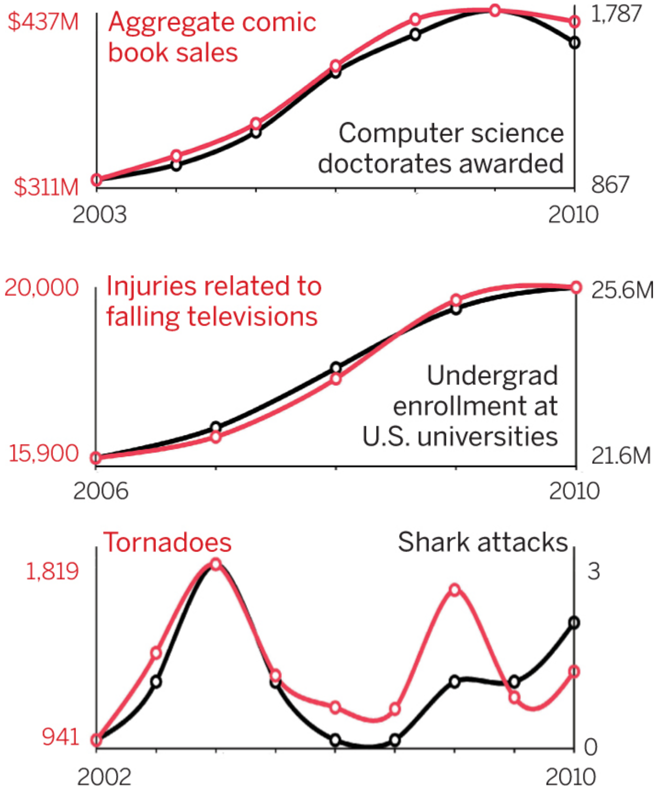
# Visualization and Correlation Pitfalls

| Pitfall | Description | How to Avoid |
|---|---|---|
| **Misinterpreting Correlation (Spurious Correlation)** | If a high correlation coefficient is there between two variables means one causes the other (causation vs. correlation). | Use domain knowledge and causality testing. Remember correlation only measures linear association. |
| **Ignoring Multicollinearity** | Using two or more highly correlated features (e.g., TotalSF and GrLivArea) in a Linear Regression model. This leads to unstable, unreliable, and uninterpretable coefficients. | Use a heatmap to visualize correlations. For highly correlated features |
| **Superficial Visualization** | Only checking scatter plots for linear relationships. Never treat EDA as just visualization. | Always use diverse visualizations: Box plots (for categorical vs. numerical), Pair Plots (for many-to-many), and Histograms (for distribution). |

# Misinterpreting Correlation as Causation




**Pitfall**

Strong correlation is interpreted as importance or causality

**Why this is dangerous**

Correlation may be:

- Proxy-driven
- Confounded
- Coincidental

# Correlation:

**Definition:** Correlation refers to a statistical relationship between two variables where changes in one variable are associated with changes in another variable. It does not imply causation.

**Example:** Ice Cream Sales and Drowning Incidents

- **Observation:** There is a positive correlation between ice cream sales and drowning incidents.
- **Explanation:** During the summer, both ice cream sales and drowning incidents tend to increase. However, it would be incorrect to conclude that buying more ice cream causes an increase in drowning incidents. The correlation is coincidental, and a third variable, such as temperature, influences both.

# Strange Correlations:

**Example:** Number of Firefighters and Property Damage

- **Observation:** There is a positive correlation between the number of firefighters at a fire station and the amount of property damage in a city.
- **Explanation:** It might be observed that cities with more firefighters tend to experience higher property damage. However, this correlation does not imply causation. A potential confounding variable could be city size. Larger cities may have both more property to protect and more resources to hire additional firefighters. The correlation is not indicative of firefighters are causing property damage.

# Causation:

**Definition:** Causation indicates that one variable directly causes a change in another variable. Establishing causation requires more rigorous study design and evidence.

**Example:** Relationship Between Smoking and Lung Cancer

- **Observation:** There is a strong correlation between smoking and the incidence of lung cancer.
- **Explanation:** Through extensive research, it has been established that smoking causes an increased risk of developing lung cancer. In this case, there is both correlation and causation. The correlation is supported by experimental evidence and a plausible mechanism.

# Causation:

**Example:** Exercise and Weight Loss

- **Observation:** There is a negative correlation between the frequency of exercise and body weight.
- **Explanation:** Through well-designed studies and experiments, it has been established that regular exercise can lead to weight loss. In this case, there is both a correlation and causation. Increased exercise directly contributes to weight loss. The correlation is supported by evidence from interventions and controlled trials.

The relationship between exercise and weight loss is not merely coincidental; there is a plausible mechanism and experimental evidence supporting causality.

This illustrates the importance of considering the context, underlying mechanisms, and study design when interpreting relationships between variables.

1. **Nature:**
   - **Correlation:** Describes a statistical association between two variables.
   - **Causation:** Implies a direct cause-and-effect relationship between two variables.
2. **Interpretation:**
   - **Correlation:** Does not imply causation. The relationship could be coincidental or influenced by a third variable.
   - **Causation:** Implies a direct cause-and-effect relationship. Establishing causation requires more rigorous evidence and study design.
3. **Examples:**
   - **Correlation:** Ice cream sales and drowning incidents.
   - **Causation:** Smoking and lung cancer.
4. **Study Design:**
   - **Correlation:** Can be observed through observational studies or statistical analysis.
   - **Causation:** Requires controlled experiments, randomized controlled trials, or other rigorous study designs to establish a cause-and-effect relationship.

# Interesting observations

- Remember: "Correlation does not imply causation." It underscores the importance of careful interpretation and the need for additional evidence before concluding a causal relationship between variables.

https://www.tylervigen.com/spurious-correlations

# Why Paradoxes Matter

What is a paradox?

Situations where the interpretation of the data appears to contradict common sense or expectations

- Statistically correct results can be conceptually wrong
- Leads to wrong conclusions in EDA and fairness audits
- Major cause of model failure in deployment

**Paradoxes are warnings that the data is asking a deeper question.**

# Why paradoxes matter in Data Science

Paradoxes arise when:

- Data aggregation hides structure
- Confounders are ignored
- Selection bias exists

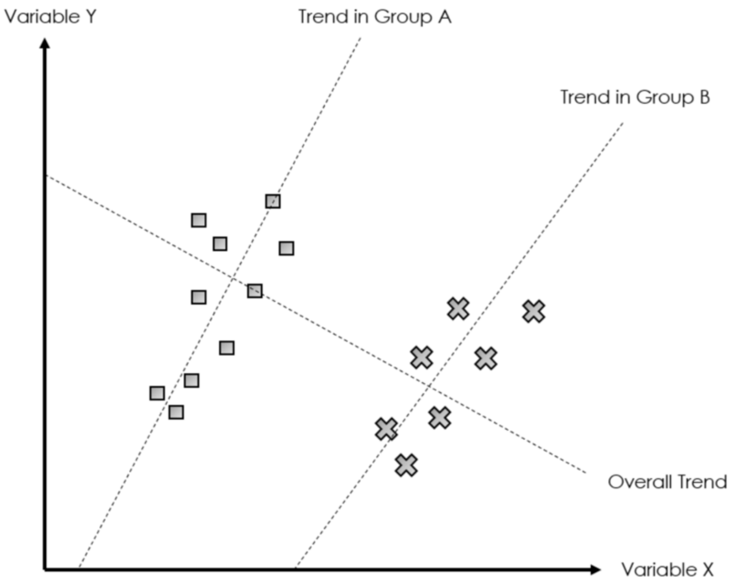
They directly impact:

- Fairness audits
- Policy decisions
- Model deployment safety

**Paradoxes arise when averages hide structure, confounders distort causality, and biased selection filters reality.**

# Data aggression hides structure

- When we combine (aggregate) data across different groups, we often average away important differences.
- At the aggregate level: Trend in group A and B looks negative
- But within each subgroup: The trend is actually the opposite
- This creates the *illusion* of a relationship that doesn't truly exist.



# Confounders are ignored

**What is a confounder?**

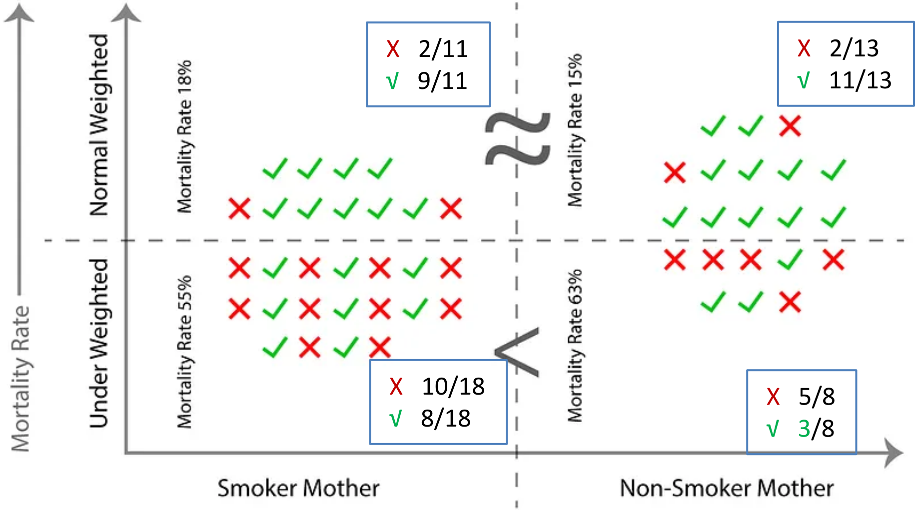
A confounder is a variable that: Influences the input (X) and influences the output (Y), but is not included in the analysis.

When a confounder is ignored:

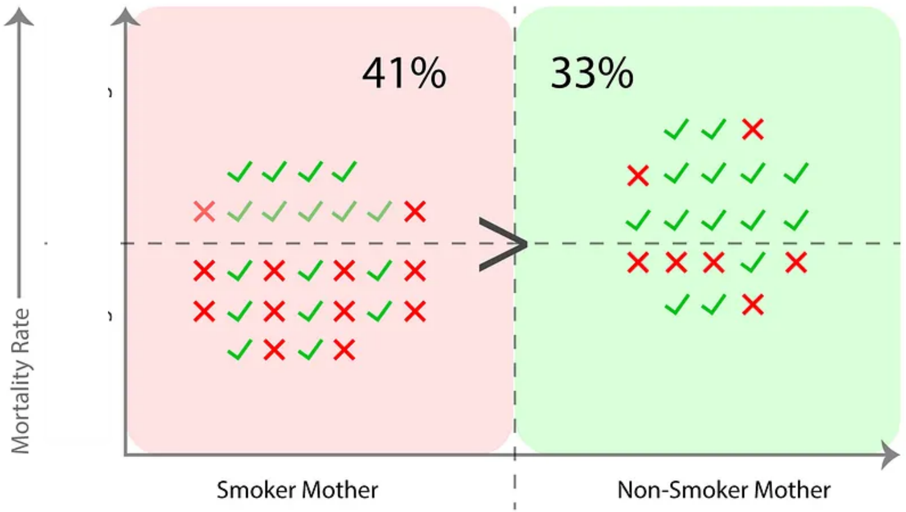
- We attribute an effect to the wrong variable
- The direction of association can flip after adjustment
- Ice cream sales ↑,Drownings ↑
- Wrong conclusion: Ice cream causes drowning
- Hidden confounder: Temperature

# Simpson's Paradox- Yule Simpsons Effect

- Data aggression hides structure
- Confounders are ignored



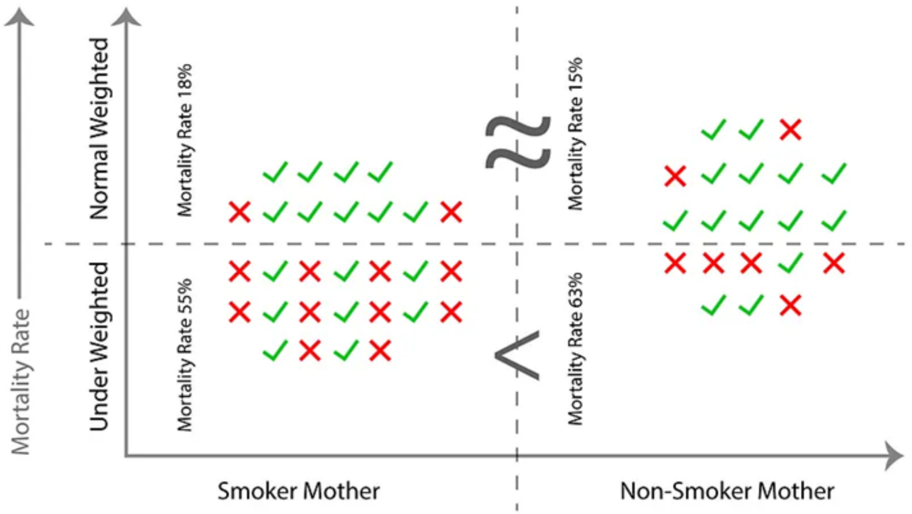

Infant mortality rate of a baby for a smoker vs non smoker mother

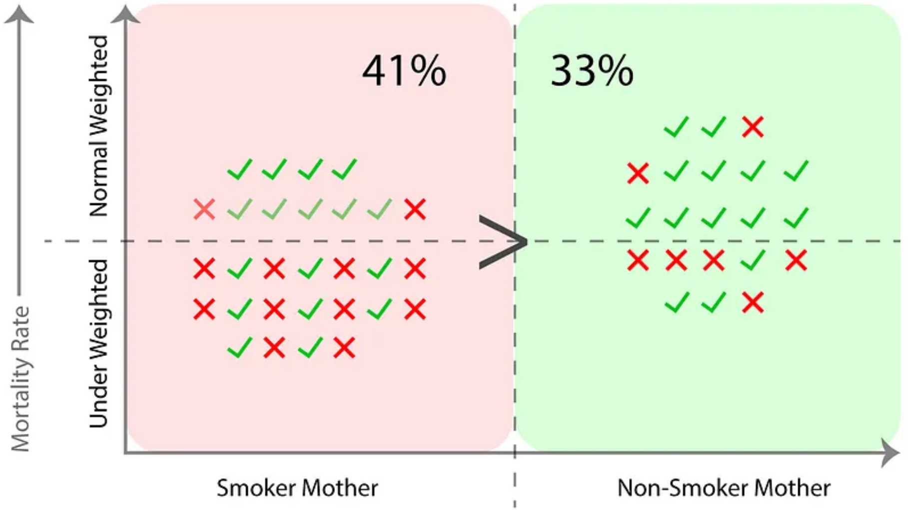


X-axis: Mother is *Smoker* vs *Non-Smoker and* Y-axis: Infant *Mortality Rate*
Hidden grouping (the confounder): Normal weight babies and under-weight babies

In under-weight babies, non-smokers mother babies appear worse (which contradicts intuition)

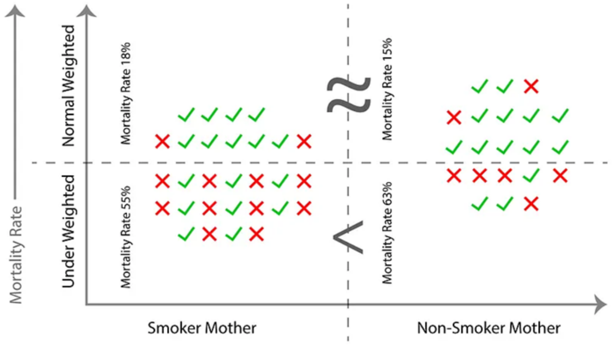
- Smoker mothers → 41% mortality for babies
- Non-smoker mothers → 33% mortality for babies
- Looks like smoking is worse (which aligns with intuition)

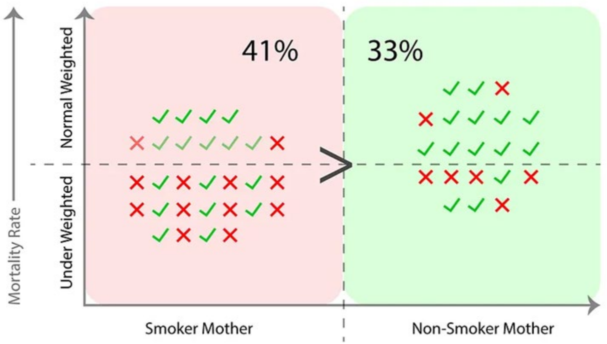


There are three facts that we can conclude from figures about infant mortality rate between smokers vs non-smokers mother's babies.

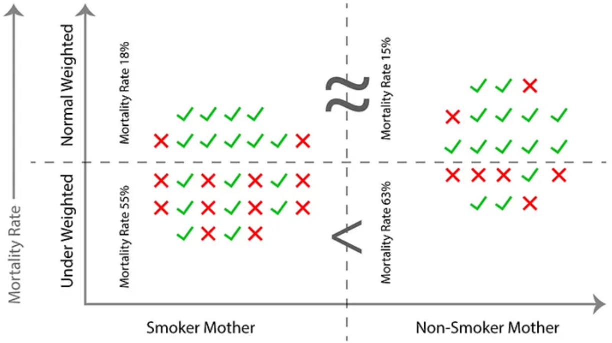
- Fact 1: Underweight smoker mother's babies have significantly less mortality rate than underweight non-smoker mother's babies. It's surprising but true according to data.
- Fact 2: Normal weight smoker mother's babies have about the same mortality rate as the Normal weight non-smoker mother's babies.
- Fact 3: If we add up both underweight and non-underweight categories you'll see that smoker mother's babies have a much higher moralities rate than non-smokers mother's babies. (causally correct)



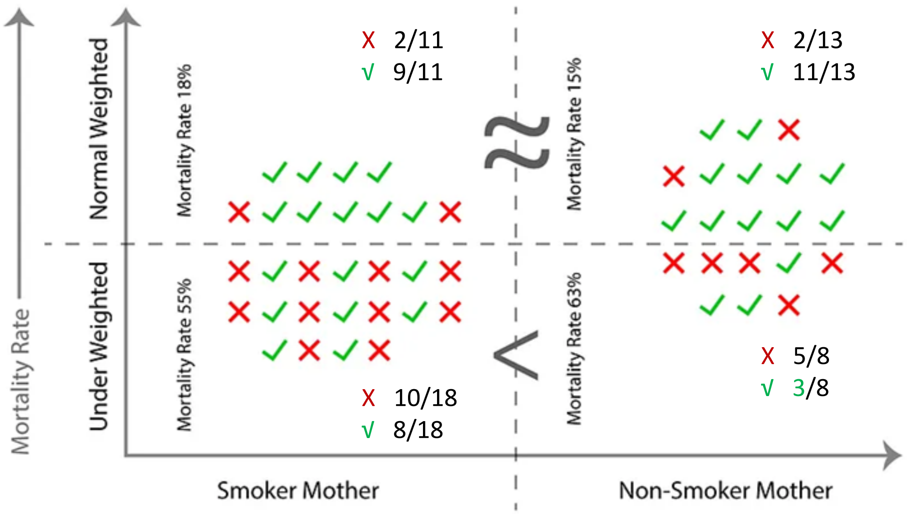
# How does knowing the confounder solve the problem?

- What we did is instead of : “Smoker vs Non-smoker”, compare:
  - Smoker vs Non-smoker within Normal weight babies
  - Smoker vs Non-smoker within Under-weight babies
- For the paradox to disappear and the comparison to become fair, we need make the confounder birthweight to be held constant
- Now the estimated effect is: **Smoking vs mortality, holding birth weight constant**




Among babies of the same birth-weight category, does smoking still increase mortality?

Or is it smoking **increases the probability of low birth weight**, and low birth weight **dramatically increases mortality?**




## Proportion (risk) of underweight babies

$$P(\text{Underweight} \mid \text{Smoker}) = \frac{18}{29} \approx 0.621$$

$$P(\text{Underweight} \mid \text{Non-smoker}) = \frac{8}{21} \approx 0.381$$

## Risk Ratio (Relative Risk)

$$RR = \frac{18/29}{8/21} = \frac{18 \times 21}{29 \times 8} = \frac{378}{232} \approx \mathbf{1.63}$$

$$\text{Mortality rate} \mid \text{Underweight} = \frac{15}{26} \approx \mathbf{57.7\%}$$

$$\text{Mortality rate} \mid \text{Normal} = \frac{4}{24} \approx \mathbf{16.7\%}$$

## Risk Ratio (Underweight vs Normal)

$$RR = \frac{15/26}{4/24} = \frac{360}{104} \approx \mathbf{3.46}$$

Number of underweighted children for a smoker mother= 18
Number of underweighted children for a non-smoker mother =8
Number of normal weighed children for a smoker mother =11
Number of normal weighed children for a non smoker mother =13

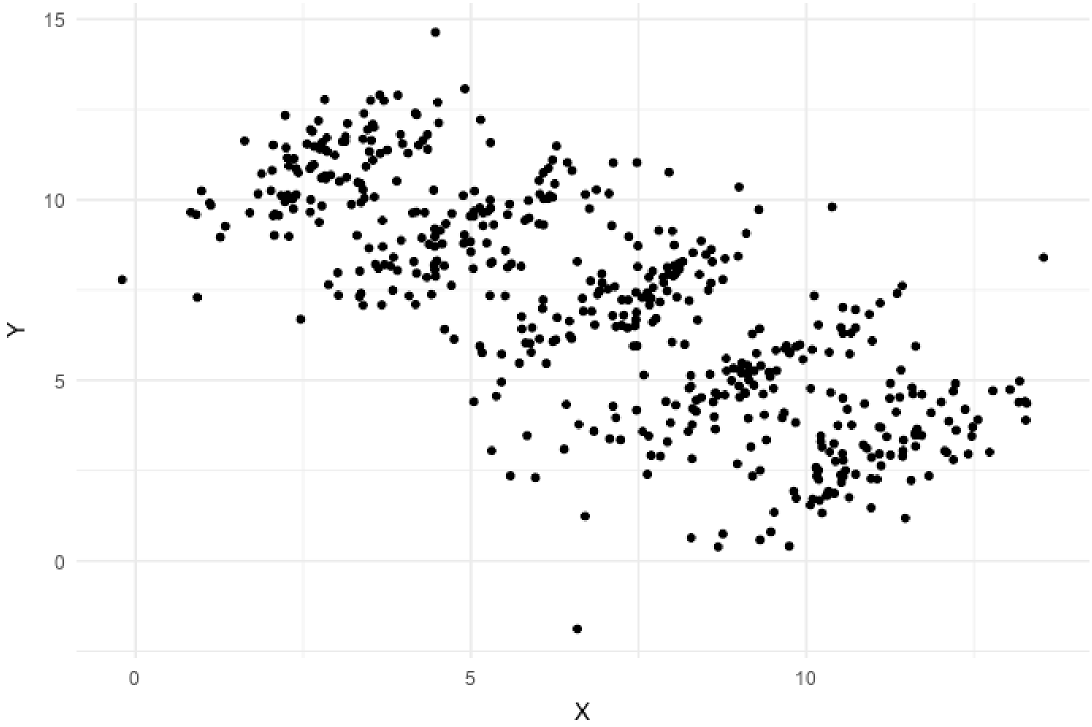
# Simpson’s Paradox- solved

**Why the contradiction happened?**

- Smokers have far more under-weight babies
- Under-weight babies have very high mortality
- Aggregation mixes two *very different risk groups*
- So the overall mortality rate is driven by: different proportions of high-risk babies, not the true causal effect of smoking.

**All three facts are statistically true, but only the stratified analysis reveals the causal mechanism; the aggregate result is confounded by birth weight, producing Simpson’s paradox.**

Korrelation:



# Why is this dangerous in Data Science

- Misleading fairness conclusions
- Incorrect feature importance
- Policy and governance failures
- Legal and ethical risks

# Berkson’s Paradox

- Selection bias exist
- Berkson’s Paradox occurs when a correlation between two variables is created or distorted because the data are conditioned on a selected subgroup.
- Example: In a population identify Disease A (heart disease) and Disease B (lung disease). Also find out the data of hospitalization for both the disease. The correlation that we cannot see in the full population, we can see in hospital data statistics as admission variable act as a collider

# Berkson's Paradox

## Full population

| Disease A | Disease B | Count |
|---|---|---|
| 0 | 0 | 7,200 |
| 1 | 0 | 1,200 |
| 0 | 1 | 1,200 |
| 1 | 1 | 400 |
| **Total** | | **10,000** |

```
Population correlation:
          Disease_A  Disease_B
Disease_A  1.000000  -0.004472
Disease_B -0.004472   1.000000
```

## If you have A or B you are admitted to hospital

| Disease A | Disease B | Count |
|---|---|---|
| 1 | 0 | 1,200 |
| 0 | 1 | 1,200 |
| 1 | 1 | 400 |
| **Total** | | **2,800** |

```
Hospital-only correlation:
          Disease_A  Disease_B
Disease_A   1.00000   -0.84394
Disease_B  -0.84394    1.00000
```

# Berkson’s Paradox

- Take away point from the correlation analysis of hospital data : If you have Disease A, you are *less likely* to have Disease B.
- This correlation is not seen in the full population
- Thus selection bias occurs due to a collider/confunder
- Berkson’s Paradox arises when conditioning on a selection variable (such as hospital admission) induces a spurious correlation between independent variables.

**Simpson’s paradox comes from averaging across groups.**
**Berkson’s paradox comes from selecting/filtering into groups.**

# Accuracy Paradox

- Data aggregation hides structure. (due to imbalanced data)
- The accuracy paradox occurs when a classification model achieves high overall accuracy but performs poorly on the classes that matter most, typically due to severe class imbalance.
- Accuracy is an aggregate metric, the majority class dominates the calculation and errors in the minority class are masked.
- Example: Dataset: 99% legitimate, 1% fraud. The model predicts “legitimate” for all cases and the accuracy = 99%. Fraud detection = 0% . This is a model with high accuracy but a useless model
- Use Recall, F1 score to understand and apply class weighting or rebalancing.

# False prediction Paradox

- Data aggregation hides structure.
- The False Prediction Paradox occurs when a model's predictions are mostly wrong for the class of interest, even though the model appears statistically reasonable or well-calibrated overall
- The model predicts many positives and most of those predictions are false. Yet the model may still look acceptable by aggregate metrics. This paradox focuses on prediction-level error, not overall accuracy.
- Example: Data set: Medical screening for a rare disease, Disease prevalence: 1% (out of 10000 people 9900 doesn't have disease and only 100 has).

out of 10000 people 9900 doesn't have disease and only 100 has

| Aspect | Accuracy Paradox | False Prediction Paradox |
| --- | --- | --- |
| Focus | Overall model accuracy | Correctness of predicted positives |
| Main symptom | High accuracy, useless model | Many predictions are false |
| Typical metric failure | Accuracy | Precision : When the model predicts SAFE, how often is it actually SAFE? |
| Root issue | Majority class dominates metric | Base-rate effect |
| Key question | "Is the model good overall?" | "Can I trust a positive prediction?" |

# Ecological Fallacy

- Ecological fallacy happens when we look at group-level (aggregated) data (e.g., countries, schools, neighbourhoods) and then make conclusions about individual-level behaviour.
- A relationship observed between groups is incorrectly assumed to hold within groups or for individuals.

| Aspect | Ecological Fallacy | Simpson's Paradox |
|---|---|---|
| **Core mistake** | Group → individual inference | Aggregated trend reverses when conditioned |
| **Data levels** | Group vs individual | Same individuals, different strata |
| **reversal** | Interpretational reversal | Yes (reversal is key) |
| **Confounder role** | Often implicit | Explicit stratifying variable |
| **Typical question** | "Can we say this about individuals?" | "Why did the trend flip?" |

# How to Detect Paradoxes in EDA

- Always compare aggregated vs stratified results
- Check subgroup metrics
- Understand data collection process- Ex: Berkson's
- Ask: What am I averaging over?

# Compare aggregated vs stratified results

- **Simpson's Paradox:**
  - **What flips**: Direction of association
  - Reversal after stratification = Simpson's
- **Ecological Fallacy:**
  - **What breaks**: Level of inference
  - Group trend ≠ individual truth
- **Accuracy Paradox:**
  - **What misleads**: Overall metric
  - Aggregate accuracy hides subgroup failure (minority has poor performance)
- **False Prediction Paradox:**
  - **What misleads**: Predictive values
  - Aggregate recall hides prediction quality (recall may be high but still false prediction)
- **Berkson's Paradox:**
  - **What flips**: Correlation sign
  - Conditioning introduces a false relationship (As we are only considering a subgroup)

# Check subgroup metrics

- **Simpson's**
  - Compare slopes / odds ratios **per subgroup**
  - If all subgroups agree but aggregate disagrees → paradox
- **Ecological Fallacy**
  - Compare: Group averages and Individual scatter within groups
  - If individual relationships differ → fallacy risk
- **Accuracy Paradox**
  - Check: Precision / Recall per class and Confusion matrix by class
  - High accuracy + low minority recall = paradox
- **False Prediction Paradox**
  - Check: False positives  Positive Predictive Value or Negative Predictive value
  - Even with high recall, predictions may be unreliable
- **Berkson's**
  - Check correlations: Full population along with Selected / conditioned sample
  - If correlation appears only after selection → Berkson's

# What am I averaging over?

- Simpson’s Paradox: You’re averaging over groups with different baselines.
- Ecological Fallacy: You’re averaging individuals into groups and then reasoning backward.
- Berkson’s Paradox: You’re averaging after conditioning on selection.
- Accuracy Paradox: You’re averaging correct predictions across imbalanced data
- False Prediction Paradox: You’re averaging detection ability without checking prediction quality.

| Paradox | What goes wrong | What aggregation hides |
| --- | --- | --- |
| Simpson’s | Direction flips | Confounder distribution |
| Ecological Fallacy | Wrong inference level | Individual heterogeneity |
| Berkson’s | Fake correlation | Selection mechanism |
| Accuracy Paradox | Misleading performance | Class imbalance |
| False Prediction Paradox | Poor decision utility | False positives/negatives |

# Key Takeaway

- Paradoxes are not errors — they are warnings
- They indicate hidden structure in data
- Always condition before concluding
- EDA must be causal, not just visual